

Linear Systems

Remember?? (From DS120)

$$\begin{aligned}x_1 - 2x_2 + x_3 &= 5 \\2x_2 - 8x_3 &= -4 \\6x_1 + 5x_2 + 9x_3 &= -4\end{aligned}$$

What we see above is an example of *linear system*. A system containing *linear equations* i.e. equations where the *highest power* of a variable is 1, no variable has *negative or fractional powers* and no two (or more) variables are *multiplied* to one another.

In other words, the only operations applied to our unknown variables are:

- (1) addition, and
- (2) multiplication by a constant.

Definitions:

A *linear equation* in variables $x_1, x_2, \dots, x_d$ is an equation that can be written in the form

$$a_1x_1 + a_2x_2 + \dots + a_dx_d = b$$

Where b and the coefficients $a_1, a_2, \dots, a_d$ are real numbers.

A *system of linear equations* (or linear system) is a collection of one or more linear equations involving the same variables, say $x_1, x_2, \dots, x_d$

Solutions, solution sets, consistent and inconsistent systems

Definitions

A ***solution of a linear system*** is a list of numbers $(s_1, s_2, \dots, s_d)$ that makes each equation a true statement when the values $s_1, s_2, \dots, s_d$ are substituted for $x_1, x_2, \dots, x_d$, respectively.

The set of all possible solutions is called the ***solution set*** of the linear system.

Two linear systems are called equivalent (***equivalent systems***) if they have the same solution set.

A system of linear equations has 0, 1, or infinitely many solutions.

A system of linear equations is said to be ***consistent*** if it has either ***one solution or infinitely many*** solutions.

A system of linear equations is said to be ***inconsistent*** if it has ***no solution***.

From equations to graphs

How do we plot a linear equation like

$$x_1 - 2x_2 = -1$$

Key idea:

Rearrange to solve for one variable in terms of the other.

$$x_1 - 2x_2 = -1 \quad \Rightarrow \quad x_2 = \frac{x_1 + 1}{2}$$

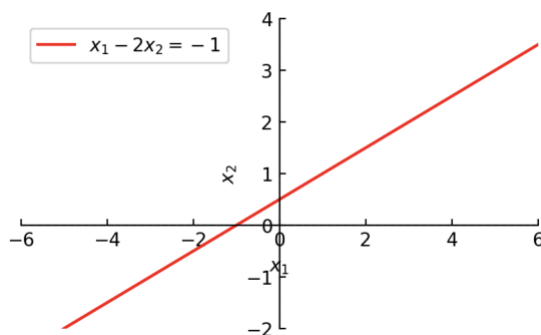
Now we can plot x_2 as a function of x_1 : and that's just a line!

In general: A linear equation in **2 variables** defines a **line** in $\mathbb{R}^2$.

A linear equation in **3 variables** defines a **plane** in $\mathbb{R}^3$.

Geometrically, we can find the solutions to a system of linear equations in d variables by:

- creating a graph showing the solution space to each individual equation, which is a $d - 1$ dimensional space;



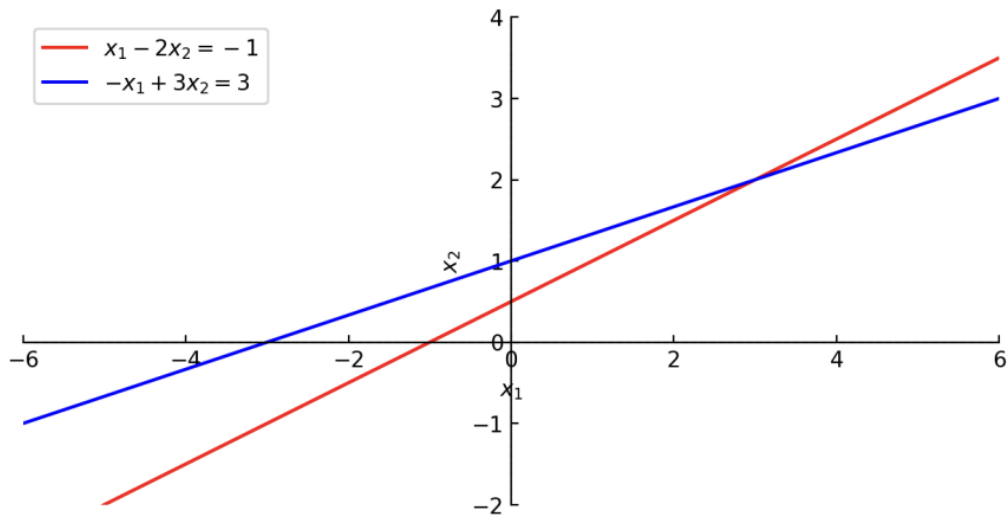
- finding the point(s) that are in common to all equations.

One solution: Intersecting lines

Let the system of equations be:

$$x_1 - 2x_2 = -1$$

$$-x_1 + 3x_2 = 3$$



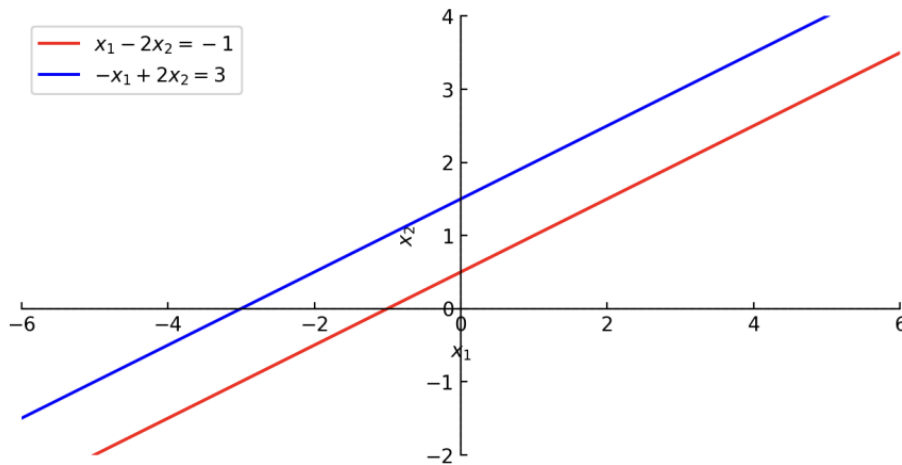
How many solutions does this system have?

No solution: Parallel lines

Let the system of equations be:

$$x_1 - 2x_2 = -1$$

$$-x_1 + 2x_2 = 3$$



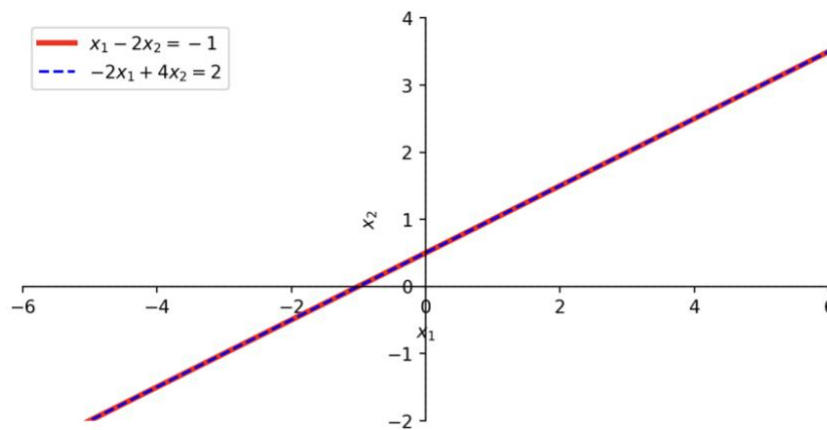
How many solutions do we have now?

Infinitely many solutions: Same line

What if the second equation is just a multiple of the first?

$$x_1 - 2x_2 = -1$$

$$-2x_1 + 4x_2 = 2$$



What is happening here?

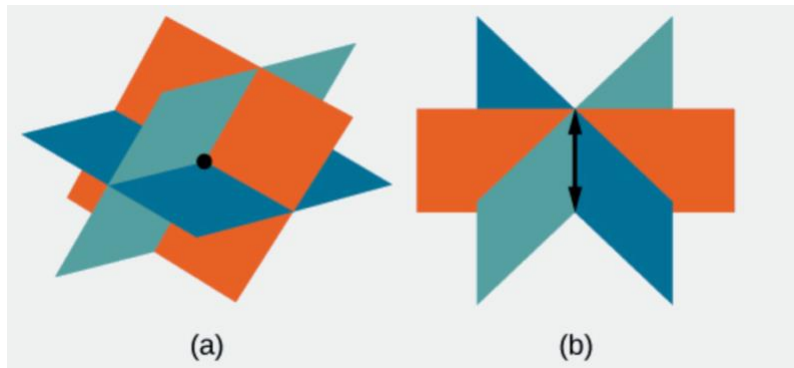
The lines overlap completely — every point on the line is a solution!

Systems in 3 dimensions

This geometric reasoning also works in higher dimensions.

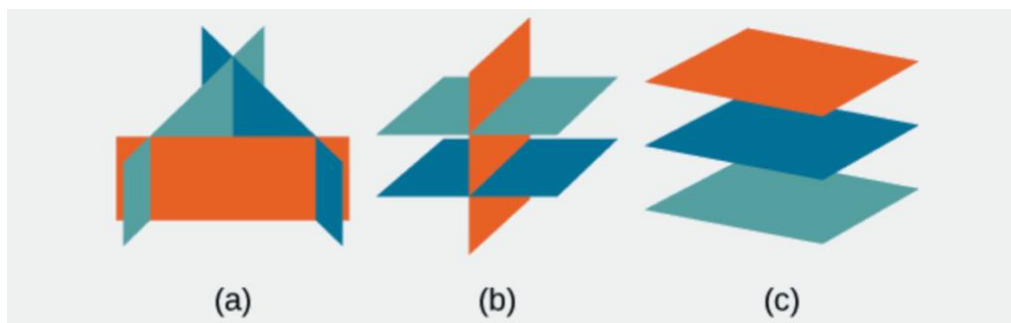
Let's consider $\mathbb{R}^3$. In three-dimensional space, the solution to each individual linear equation forms a 2-dimensional plane.

Consistent systems:



(a) Three planes intersect at a ***single point***, representing a three-by-three system with a ***single*** solution. (b) Three planes intersect in a ***line***, representing a three-by-three system with ***infinite solutions***.

Inconsistent systems:



All three figures represent three-by-three systems with ***no solution***. (a) The three planes ***intersect*** with each other, but ***not at a common point***. (b) ***Two*** of the planes are ***parallel*** and ***intersect*** with the ***third plane***, but not with each other. (c) All ***three planes*** are ***parallel***, so there is no point of intersection.

Vector equations

What happens if we group terms by variable instead of by equation? Notice that:

$$\begin{array}{rcl} x_1 - 2x_2 + x_3 & = & 5 \\ 2x_2 - 8x_3 & = & -4 \\ 6x_1 + 5x_2 + 9x_3 & = & -4 \end{array} \Leftrightarrow x_1 \begin{bmatrix} 1 \\ 0 \\ 6 \end{bmatrix} + x_2 \begin{bmatrix} -2 \\ 2 \\ 5 \end{bmatrix} + x_3 \begin{bmatrix} 1 \\ -8 \\ 9 \end{bmatrix} = \begin{bmatrix} 5 \\ -4 \\ -4 \end{bmatrix}$$

The equation on the right is called a ***vector equation***.

It involves ***column vectors*** and is equivalent to the system on the left, where ***linear equations*** arranged in ***rows***.

Let us try to connect the dots from system of equations to vector equations!

Span

Vector equations can also be inconsistent. When does this happen?

Let's take the example above.

$$\text{Let } \mathbf{a}_1 = \begin{bmatrix} 1 \\ 0 \\ 6 \end{bmatrix}, \mathbf{a}_2 = \begin{bmatrix} -2 \\ 2 \\ 5 \end{bmatrix}, \mathbf{a}_3 = \begin{bmatrix} 1 \\ -8 \\ 9 \end{bmatrix}, \mathbf{b} = \begin{bmatrix} 5 \\ -4 \\ -4 \end{bmatrix}$$

We need to determine if $\mathbf{b}$ can be written as a *linear combination* of $\mathbf{a}_1, \mathbf{a}_2$ and $\mathbf{a}_3$

That is, if there exists x_1, x_2, x_3 such that $x_1\mathbf{a}_1 + x_2\mathbf{a}_2 + x_3\mathbf{a}_3 = \mathbf{b}$.

In other words, we want to find if $\mathbf{b}$ is in the *span* of $\{\mathbf{a}_1, \mathbf{a}_2, \mathbf{a}_3\}$.

If no such x_1, x_2, x_3 , then $\mathbf{b}$ is *not* in the *span* of $\{\mathbf{a}_1, \mathbf{a}_2, \mathbf{a}_3\}$ and the system is *inconsistent*.

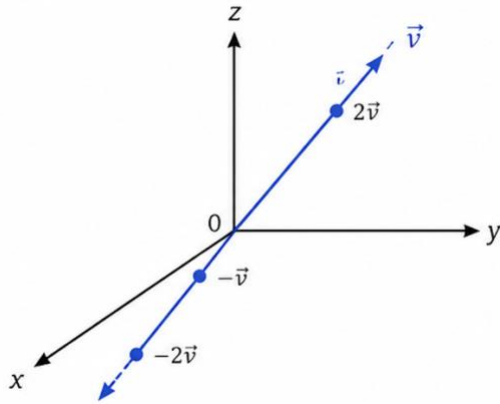
Definition

Consider a set of vectors $\mathbf{a}_1, \mathbf{a}_2, \dots, \mathbf{a}_k$ where each $\mathbf{a}_i \in \mathbb{R}^d$. Then the set of all linear combinations of $\mathbf{a}_1, \mathbf{a}_2, \dots, \mathbf{a}_k$ is denoted by $\text{Span}\{\mathbf{a}_1, \mathbf{a}_2, \dots, \mathbf{a}_k\}$ and is called the *subset of $\mathbb{R}^d$ spanned by $\mathbf{a}_1, \mathbf{a}_2, \dots, \mathbf{a}_k$* .

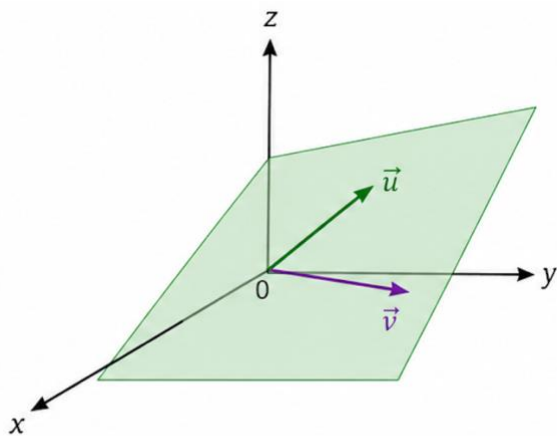
What is the span of a single non-zero vector?

Span of vectors in $\mathbb{R}^3$

Span of single vector in $\mathbb{R}^3$



Span of two vectors in $\mathbb{R}^3$



What if one vector is in the span of the other?

What about 3 vectors in $\mathbb{R}^3$?

Two distinct perspectives

Be sure to keep clear in your mind that we have been working with two different vector perspectives:

1. **Row view:** Each equation defines a line/plane. Find their intersection.
2. **Column view:** Can we reach b using linear combinations of column vectors?

Example (Really a metaphor!)

Suppose you have two tubes of paint:

$yellow = \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix}$ and $cyan = \begin{bmatrix} 0 \\ 1 \\ 1 \end{bmatrix}$ with channels $\begin{bmatrix} R \\ G \\ B \end{bmatrix}$. Can you mix them to get

$pure\ red = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}$?

- **Column view:** Any combination of yellow and cyan will have some green or blue in it. Pure red has no green or blue – it's not in the span.
- **Row view:** For red, the R channel need $x_1 = 1$ and B channel needs $x_2 = 0$, but then the G channel requires $x_1 + x_2 = 0$ which is a contradiction!

Same conclusion but two framing: ***mixing ingredients*** vs ***satisfying constraints!***

For each system, determine whether it has one solution, no solutions or infinitely many solutions

- $x + y = 3$
 $2x + 2y = 6$

- $x + y = 3$
 $x + y = 5$

- $x + y = 3$
 $x - y = 1$